

Real-time video anomaly detection — 11 classes, open-set

One frozen encoder, encoded once, read three ways. No backbone fine-tuning.

58.8

benchmark score

18–58×

realtime

125 s

all 28 videos

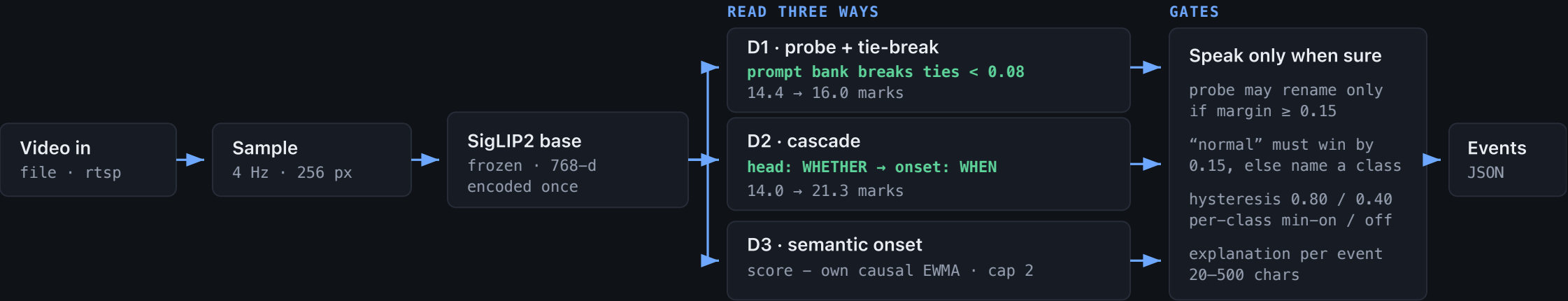
2.4 MB

trained weights

A2000

12 GB GPU

PIPELINE



WHY THIS SHAPE

- **Frozen encoder, one pass.** Features computed once, read three ways — why it beats realtime on a modest GPU.
- **Prompts are data, not weights.** A twelfth class is a sentence, not a retrain.
- **Score against normal.** Margin over the best of ~70 normal prompts; raw similarity just describes the scene.
- **Long video needs a baseline.** D3 keys on the rise over a causal 20 s EWMA, so a busy junction isn't permanently anomalous.
- **Locating ≠ naming.** Detector says when, probe says what — 63% vs 33% on class.

TESTED AND REJECTED

TRIED	OUTCOME
Half-max boundary refinement	D3 16.9 → 8.0
Adaptive per-video D2 thresholds	D2 14.0 → 5.3

Nine full runs. Every gain was routing; every loss was tuning.

Each row is one complete pipeline run over all 28 videos, scored live. Nothing estimated, nothing assembled.

EVERY SUBMISSION, MEASURED

#	WHAT CHANGED	D1	D2	D3	MARKS
1	Baseline: probe · head · onset	11.6	14.0	16.9	42.5
2	“normal” must win by a margin	14.4	14.0	16.9	45.3
3	Half-max boundary refinement	14.4	14.0	8.0	36.4
4	Onset instead of head on D2	14.4	12.6	16.9	43.9
5	Adaptive per-video D2 thresholds	14.4	5.3	16.9	36.6
6	Tighter D2 hysteresis	14.4	14.0	16.9	45.3
7	D2 cascade: whether → when	14.4	21.3	16.9	52.6
8	Keep 2 D3 intervals, not 3	14.4	21.3	17.9	53.6
9	Prompt bank breaks probe ties	16.0	21.3	17.9	55.3

+3.5/5 reasoning bonus throughout — **58.8 total**. Routing changes: +2.8, +7.3, +1.0, +1.7. Fitted parameters: -8.9, -8.7, -1.4.

THE TWO FAILURES THAT TAUGHT US THE MOST

- **Run 3 (D3 halved)**. We shrank intervals to their peak width. D3's real events run 38–125 s, so a tidy span misses them entirely. Lesson: cap the *count*, don't trim the *extent* — which became run 8.
- **Run 5 (D2 collapsed)**. Forcing every video to fire cost 8.7 ≈ exactly one video (8.75). That accident revealed run 7: the head's *silence* on normal footage was worth more than its localisation.

THE CASCADE, FROM RUNS 4–6

	NORMAL VIDEO	3 ANOMALOUS	D2
head only	8.75	5.25	14.0
onset only	0	12.6	12.6
cascade	8.75	12.6	21.3

Neither component could do both jobs. Chaining them collects both.

WHAT ACTUALLY FOUND THE BUGS

- **Looking at the frames**. One labelled contact sheet of every predicted event. Ten minutes, four real defects.
- A clip called `wrong_way_driving` ×6 — it was queued trucks on the far carriageway.
- A dashcam **crash compilation** typed `stalled_vehicle` — watermark visible in shot.
- Run 9 then flipped `fire`↔`smoke` on **exactly the two clips we'd flagged by eye**, from the prompt bank alone.

LOCAL VALIDATION LIED — FIVE TIMES

SIGNAL	SAID	TRUTH
Marks formula	29.9	51.9
Train accuracy	+27	-24
Val IoU 0.66→0.86	better	worse
Cross-val 33→58%	better	±0

Ground truth was revised mid-event; every cached label was stale. Pixels don't go stale.